

A consistent price, tax and trade block

Design spec for LEEDS_MODEL — written 2026-09-11, awaiting review

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Status: design approved in conversation 2026-09-11; this written spec awaits the maintainer’s review. **Branch at writing:** `main @ 094fef0` (after the `bop-audit` merge `65f2985`). **Implementation plan:** to follow (`/plan`), not yet written.

1. Why this exists

The model’s price of production is classical and internally sound: intermediates enter at the VAT-exclusive, duty-free natural price `p_t` (`model/MVP_model_2026.R:602–608`). Everything downstream of it — the market price, the deflators, the trade account, the tax revenue lines — was found on 2026-09-11 to be inconsistent with it and with itself. Measured on the cached `customs`-arm baseline (`output/scenarios/arms/customs/runs/baseline.RDS`), `t = 99`, model currency units per period; scripts in `plans/2026-09-11-evidence/` (`tax_consistency.R`, `tax_consistency2.R`, `import_channels.R`, `gamma_gap.R`, `foreign_share.R`; read-only, run with `Rscript <file>`):

#	Inconsistency	Line	Measured (EU / RoW)
a	The import price index <code>pim</code> applies the partner’s tariff rate to the exporter’s VAT-inclusive market price and subtracts the own VAT additively. It is wrong under either tariff incidence.	<code>:646</code>	<code>pim</code> exceeds the duty-free, VAT-free price the importer pays by 12.0% / 7.6% (1.0147 vs 0.9059 EU)
b	VAT charged but never collected on investment, government investment and government consumption: their deflators use the VAT-inclusive <code>p</code> ; <code>vat_rev</code> taxes household consumption only; the VAT stays in <code>yn</code> and becomes firm profit.	<code>:405</code> , <code>:628–643</code> , <code>:215–223</code>	69.4 / 260.3 , 6.15% of <code>yn</code> . The EU government pays firms 42.0 of VAT a period, more than its whole primary surplus (≈ 29 , 2.3% of <code>va</code>)

#	Inconsistency	Line	Measured (EU / RoW)
c	<code>va</code> is valued at VAT-inclusive prices on both sides, so the debt ratio's denominator is inflated by <code>1 + vat</code> .	:235–237	EU <code>debt_gdp</code> 0.8807 reported vs 1.0148 at basic prices (canonical baseline; <code>log/report/2026-09-09-tax-wedge-stage.qmd</code>)
d	Two final-import volumes. Production (<code>x = L · d</code> , <code>d</code> at :184–193) uses each agent's expenditure shares over all 108 goods; the trade account, the tariff base and the partner's exports use an aggregate import function <code>imp = exp(mu0 - mu1 · log(pim/pa) + mu2 · log(yd/pa))</code> (:756). The commented-out older <code>d</code> at :195–202 subtracted <code>imp</code> and so had one volume; the full-MRIO <code>d</code> did not retire the import function.	:184–193, :756	real final imports 61.16 (import function) vs 67.95 (Leontief system), –10.0%; RoW 103.23 vs 106.84, –3.4%. Accounts for 2.82 of a 2.88 mirror residual between income and production. In Sc5 and Sc14 the two volumes move in opposite directions (EU, $t = 90$: +0.040% vs –0.016%; +0.023% vs –0.011%)
e	VAT is priced by origin (one <code>p</code> vector carrying the producer region's VAT) but collected by destination (buyer's rate).	:616, :405	0.04 / –0.06 — negligible, rates differ by 0.0012

Not inconsistencies, but facts the design rests on:

- The market-price gap `p / ((1 + vat) · p_t) - 1` — the demand-pressure term `y(x - x*)` — is below **4.2×10^{-6}** of the natural price after $t = 50$, in the baseline and in Sc5, Sc10, Sc13, Sc14.
- In the 14 scenarios producer prices move by at most **0.014%** (Sc14, EU, $t = 90$), and the import function's price term moves final imports by at most **0.010%** (Sc10, RoW). The income term `mu2` carries most of today's import response.
- `mu0`, `mu1`, `mu2` have **no recorded provenance** — not in the supplement, the revision, `ai_context/` or the logs; the workbook (`data/full_mrrio_initial_state.xlsx`, sheet `aggregate.pars`) labels them only “Parameter in Import Equation”.

2. Decisions (maintainer, 2026-09-11)

1. **Tariff base** `customs`, **exporter incidence** as the default. (Already applied to the revision's text at `094fef0`.)
2. **The Leontief system's cross-border blocks are the single import volume.** The aggregate import function is retired.
3. **An Armington origin split, disabled by default** (elasticity 0 $\Rightarrow$ fixed shares).
4. **Standard VAT.** Households pay their own region's VAT on every good; firms' investment and government purchases are at basic prices.
5. **A tariff pass-through share, default 0** (full exporter absorption).
6. **Build behind a switch, run one baseline, then decide on recalibration.**

3. The design

3.1 One chain of prices

- **Natural price** `p_t` (`:602–608`): unchanged.
- **Market price at basic values** $p = p_t + \gamma_x \odot (x - x_{-1})$. Under the new block the `p` row *means* this price — no VAT. Every transaction is valued at it: gross output value `y` (`:212`), value added `va_j` (`:235–237`), the real value-added index (`:243–245`), target capital (`:263–268`). `va` and hence `debt_gdp` become basic-price measures (fixes c).
- **Buyer prices**, for an agent of region z buying good s from origin o :

$$q_{o,s}^{hh,z} = \text{CIF}_{o,s}^z (1 + \text{vat}_s^z), \quad q_{o,s}^{a,z} = \text{CIF}_{o,s}^z \quad (a = \text{firm investment, government})$$

where `CIF` is the duty-inclusive price the importing buyer pays (3.2), equal to `p` for a domestic good. The deflators `pa`, `pid`, `pid_g`, `pg` are the share-weighted averages of these buyer prices (fixes b and e).

3.2 Tariffs, with a pass-through share

Let $\varphi^z \in [0, 1]$ be region z 's pass-through share. For good s imported by z from f :

$$\text{CIF}_s^z = x r^f p_s^f (1 + \varphi^z \tau_s^z), \quad \text{CV}_s^z = \frac{\text{CIF}_s^z}{1 + \tau_s^z}, \quad \text{duty}_s^z = \tau_s^z \text{CV}_s^z$$

`CV` is the customs value — what the exporter receives, net of duty. The importing buyer pays $\text{CIF} = \text{CV}(1 + \tau)$ and remits the duty to its own government.

- $\varphi = 0$ (default): $\text{CIF} = x r \cdot p^f$, $\text{CV} = x r \cdot p^f / (1 + \tau)$, $\text{duty} = x r \cdot p^f \cdot \tau / (1 + \tau)$ — exactly the `customs` arm; the exporter bears the whole duty through a lower realised export price.
- $\varphi = 1$: $\text{CV} = x r \cdot p^f$, $\text{CIF} = x r \cdot p^f (1 + \tau)$, $\text{duty} = x r \cdot p^f \cdot \tau$ — importer incidence.

Trade is valued at the customs value (national-accounts standard: imports and exports at the border, duties as a tax on products paid by the importer):

$$nimp^z = \sum_s CV_s^z m_s^z, \quad nex^z = xr^z nimp^f, \quad TAR^z = \sum_s \tau_s^z CV_s^z m_s^z$$

with m^z_s the real cross-border volume of good s (3.3). National income becomes

$$yn^z = c^z pa^z + id^z pid^z + id_g^z pid_g^z + g^z pg^z + nex^z - nimp^z - VAT^z - TAR^z.$$

Equivalence with the bop-audit settlement line. At $\varphi = 0$, $xr \cdot p^f \cdot m - CV \cdot m = TAR$, so the trade balance at customs value equals the trade balance at $xr \cdot p^f$ plus $TAR^z - xr^f \cdot TAR^f$ — the net tariff settlement 865749e adds at :811–813. The two formulations give the same current account. **Under the new block the settlement line must be skipped**, or the settlement is counted twice; yn deducts the *own* duty (paid by the importing buyer) instead of the partner's.

Consequence of $\varphi = 0$: no buyer's price contains a duty, so a tariff is a pure transfer from exporters' margins to the importing government and has no relative-price effect. No current scenario varies a tariff; $\varphi > 0$ exists for later tariff or border-carbon scenarios.

3.3 Trade volumes

- **Real imports by good** m^z_s = intermediate block (M_TOT_int , :740–744, already computed from A and x) + the cross-border block of each of z 's agents' final demand ($c \cdot \beta$, $id \cdot \iota$, $id_g \cdot \iota_g$, $g \cdot \sigma$, restricted to the partner's 54 goods). By good, not aggregated with η weights. Fixes d.
- **Rows kept, meanings stated:** imp = real final imports from the Leontief system; $tot_imp = imp + M_TOT_int$; $nimp$, nex as 3.2; $pim = nimp / tot_imp$, a derived unit value; rex = partner's imp (read by nothing today). The import function and μ_0 , μ_1 , μ_2 are not read under the new block.
- **What is lost:** $\mu_2 = 0.447$ made the import propensity fall with income. Fixed or Armington shares are homothetic, so imports move one-for-one with each demand component. This is retired, not relocated.

3.4 The Armington origin split

The framework. Goods of the same industry from different regions are imperfect substitutes; each buyer splits its spending on industry s between origins according to relative buyer prices with a constant elasticity of substitution σ (Armington 1969). Here it acts on the expenditure shares β , ι , ι_g , σ (the model's government shares; the elasticity is written ε below to avoid the clash).

For agent a of region z , industry s , origins $o \in \{Z1, Z2\}$, with $\alpha_{\{o,s\}}$ the shares **after** `compute_all_delta_eff` has applied the scenario:

$$w_{o,s,t} = \left(\sum_{o'} \alpha_{o',s} \right) \frac{\alpha_{o,s} (q_{o,s,t-1}/q_{o,s,1})^{-\varepsilon^z}}{\sum_{o'} \alpha_{o',s} (q_{o',s,t-1}/q_{o',s,1})^{-\varepsilon^z}}$$

- **Industry totals are preserved**, so the split cannot undo a scenario's primary→secondary substitution (which moves shares *across* industries within each origin).
- **Lagged** buyer prices, as the retired import function used; the reference is period 1 (built from the natural price p_t , whose meaning does not change, because the period-1 p row in the workbook is VAT-inclusive).
- $\varepsilon^z = 0$ **by default** $\Rightarrow w \equiv \alpha$ exactly (fixed shares).
- **Values for when it is switched on:** converted from μ_1 . For a CES split, the foreign quantity's own-price elasticity is $\varepsilon(1 - s_f)$, so $\varepsilon = \mu_1/(1 - s_f)$ with s_f the foreign share of real final demand (5.98%

EU, 2.47% RoW): $\epsilon_{Z1} = 0.626$, $\epsilon_{Z2} = 0.783$. With $\epsilon < 1$, the foreign expenditure share rises when foreign goods become dearer. ⚠️ μ_1 's provenance is unknown and the plausibility of $\epsilon < 1$ for these aggregates has not been checked against the literature.

3.5 VAT

$VAT^z = \sum_s \text{vat}^z_s \cdot CIF^z_s \cdot (c^z \beta^z_s)$ over both origins — household purchases only, at the buyer's rate, on the duty-inclusive price (destination principle: VAT on customs value plus duty). Firms and governments pay no VAT (deductible; self-paid).

4. Mechanics

- **Switches:** `options(leeds.price_spec = "coded" | "consistent")`, default "coded"; `options(leeds.armington_eps = c(Z1 = 0, Z2 = 0))`; `options(leeds.tar_passthrough = c(Z1 = 0, Z2 = 0))`. "consistent" stops with an error unless `leeds.tar_spec == "customs"`.
- **Code:** pure vector functions (buyer prices, customs value, Armington shares, import volumes by good) in a new `model/price_block_2026.R`, sourced by `model/bootstrap_2026.R`. The model file calls them inside `if (.price_spec == "consistent")` branches at :184 (shares), :405–447 (revenues), :616–651 (prices), :737–778 (trade), :811 (settlement skipped), :215–223 (yn).
- **Storage:** only rows that already exist in the workbook are written; per-good volumes and buyer prices are locals.
- **Flash model:** regenerated with `python3 tools/build_flash_model.py` after every model edit; helpers take vectors, not labels, so the generator's rewrite does not touch them.

5. Acceptance

A. Switch off ("coded"): a same-session `identical()` comparison of the full baseline against the current model — the new code must change nothing by default.

B. Switch on ("consistent", customs , $\epsilon = 0$, $\phi = 0$), every period from $t = 3$, both regions, to solver tolerance, in a new `R/price_block_check.R` that exits non-zero on failure:

1. $yn^z = va^z - \sum_s (xr \cdot p^z_s - CV^f_s) \cdot m^f_s$ — income equals basic-price value added less the duty exporters absorb. (The 2.88 residual goes to zero.)
2. Real imports equal the Leontief cross-border volumes: $\text{tot_imp}^z = \sum_s m^z_s$.
3. VAT^z equals the VAT inside household purchases; no VAT inside `pid`, `pid_g`, `pg`.
4. `R/bop_audit.R` on the new baseline: CONSISTENT, **with no settlement term** — its rebuild of `cab` must be taught the customs-value valuation.
5. $\epsilon = 0 \Rightarrow$ shares identical to α ; a test value $\epsilon = 0.626 \Rightarrow \sum_o w_{\{o,s\}} = \sum_o \alpha_{\{o,s\}}$ for all s , and Sc13's substitution preserved.
6. Gauss–Seidel never hits its iteration cap; `consistency.error` no larger than today's order.

C. The decision point. One table, coded vs consistent baseline at $t = 70$ and $t = 99$: debt/VA, primary balance/VA, current account/VA, wage share `shw`, profit share `shp`, real imports, per-period growth of gross output value

(stationarity), iterations per period. The maintainer decides on recalibration from it. **Nothing beyond one baseline run is regenerated before then.**

6. Risks

- **The first periods.** The workbook's period-1 p is VAT-inclusive; :243 and :264 read $p[, i-1]$, so $t = 2$ sees the old meaning. Expect a transient; acceptance starts at $t = 3$.
- **The baseline will move** — government spending alone falls 42.0/period in the EU. The initial stocks were calibrated under the old equations; whether the new baseline settles is not known.
- **Sc5 and Sc14 current-account signs** may change, because their two import volumes move in opposite directions today (1, d).
- **The supplement's eq 2.9–2.10 and its settlement paragraph** (added at 094fef0) describe the coded block; they must be rewritten when the new block becomes the one reported.

7. Out of scope

- A carbon tax — none exists in the model. The two-primary-input form $p = (w \cdot l + p_c \cdot e) [I - A]^{-1}$ (log/report/2026-09-09-tax-wedge-stage.qmd) is a later item; the buyer-price functions are where it would attach.
- The floating exchange rate. A.xr reads xr from period 1 only (:49–59) — harmless while xr is pinned at 1; noted, not fixed.
- Recalibration, and regenerating the 45 arm runs — after decision C.
- Hand-off items 2 (regime claim) and 3 (supplement Appendix 5).